VLAN Configuration with Trunking and Native VLAN 999

Cisco Packet Tracer Project Documentation

1. Introduction

The objective of this project is to configure a small departmental network using two Cisco switches. Virtual Local Area Networks (VLANs) separate the IT, HR, and Management departments. A trunk link between the switches carries traffic for multiple VLANs over one physical connection. VLAN 999 is configured as the native VLAN.

2. Project Objectives

• Create and configure multiple VLANs.

• Assign switch ports to their respective VLANs.

• Establish trunk communication between two switches.

• Configure a dedicated native VLAN.

• Configure switch management IP addresses.

• Isolate unused switch ports.

• Verify VLAN and trunk configurations.

3. VLAN Details

VLAN ID	VLAN Name	Purpose
10	IT	IT Department
20	HR	HR Department
30	Management	Management Department
99	Unused_Ports	Unused switch ports
999	Native	Native VLAN for trunking

4. Physical Network Connections

Two Cisco 2960 switches (S1 and S2) are connected through their GigabitEthernet0/1 interfaces. FastEthernet interfaces are configured as access ports for departmental devices. Unused interfaces are assigned to VLAN 99 and administratively shut down.

• S1 G0/1 connects to S2 G0/1.

• Fa0/1–Fa0/6 are assigned to VLAN 10.

• Fa0/7–Fa0/12 are assigned to VLAN 20.

• Fa0/13–Fa0/18 are assigned to VLAN 30.

• Fa0/19–Fa0/24 and G0/2 are assigned to VLAN 99 and shut down.

5. Switch Configuration

Apply the following VLAN and access-port configuration on both switches. The interface names below assume a Cisco 2960 model with GigabitEthernet0/1 and GigabitEthernet0/2.

Step 1: Create VLANs

```
enable
```

```
configure terminal

vlan 10
 name IT
vlan 20
 name HR
vlan 30
 name Management
vlan 99
 name Unused_Ports
vlan 999
 name Native

end
```

Step 2: Configure Department Access Ports

```
configure terminal

interface range fa0/1-6
 switchport mode access
 switchport access vlan 10

interface range fa0/7-12
 switchport mode access
 switchport access vlan 20

interface range fa0/13-18
 switchport mode access
 switchport access vlan 30

end
```

Step 3: Configure and Shut Down Unused Ports

```
configure terminal
interface range fa0/19-24, gi0/2
 switchport mode access
 switchport access vlan 99
 shutdown
end
```

Step 4: Configure the Trunk Port

Apply this configuration on both S1 and S2. Both ends of the trunk must use the same native VLAN.

```
configure terminal
interface gigabitEthernet0/1
 switchport mode trunk
 switchport trunk native vlan 999
end
```

Step 5: Configure the Management SVI on S1

```
configure terminal
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
end
copy running-config startup-config
```

Step 6: Configure the Management SVI on S2

```
configure terminal
interface vlan 30
 ip address 192.168.30.2 255.255.255.0
 no shutdown
end
copy running-config startup-config
```

6. IP Address Configuration

Configure the following static IPv4 addresses on the PCs in Packet Tracer.

Device	VLAN	IP Address	Subnet Mask
IT-PC1	10	192.168.10.10	255.255.255.0

Device	VLAN	IP Address	Subnet Mask
IT-PC2	10	192.168.10.11	255.255.255.0
HR-PC1	20	192.168.20.10	255.255.255.0
HR-PC2	20	192.168.20.11	255.255.255.0
Management-PC1	30	192.168.30.10	255.255.255.0
Management-PC2	30	192.168.30.11	255.255.255.0
S1 (SVI)	30	192.168.30.1	255.255.255.0
S2 (SVI)	30	192.168.30.2	255.255.255.0

Default gateway: A default gateway is not required for communication within the same IP subnet. Communication between different VLANs requires a Layer 3 routing configuration.

7. Verification and Testing

Run the following commands on both switches to verify the configuration.

```
show vlan brief
show interfaces trunk
show ip interface brief
show running-config
```

Recommended Ping Tests

• Ping from IT-PC1 (192.168.10.10) to IT-PC2 (192.168.10.11).

• Ping from HR-PC1 (192.168.20.10) to HR-PC2 (192.168.20.11).

• Ping from Management-PC1 (192.168.30.10) to Management-PC2 (192.168.30.11).

• Ping between the switch management IP addresses: 192.168.30.1 and 192.168.30.2.

• Ping from a Management PC to each switch management IP.

Successful responses confirm connectivity within the relevant VLANs. The management SVI must be operational (up/up) for management connectivity. Its VLAN must exist and have at least one active Layer 2 port; the trunk must also be up and carry VLAN 30.

8. Important Networking Concepts

VLAN: A logical network that separates devices into different broadcast domains.

Access Port: A switch port assigned to a single VLAN, commonly used to connect end devices.

Trunk Port: A link that carries traffic from multiple VLANs between network devices using VLAN tagging.

Native VLAN: The VLAN used for untagged traffic on an IEEE 802.1Q trunk.

SVI (Switch Virtual Interface): A logical interface that provides an IP address for switch management or Layer 3 functions.

9. Technical Notes

This design uses Layer 2 switches and does not include a router or Layer 3 switch. Therefore, the IT, HR, and Management VLANs are separate networks and cannot communicate with one another without inter-VLAN routing.

For a more secure and controlled trunk, consider explicitly allowing only the required VLANs (10, 20, 30, and 999) on both trunk interfaces. Also ensure VLAN 999 is not used for end-user devices. The exact interface-range syntax should be checked against the switch model and Packet Tracer version.

10. Conclusion

This project demonstrates the practical implementation of VLAN segmentation, access port assignment, trunking, native VLAN configuration, and switch management using Cisco IOS commands. It provides hands-on experience with departmental network organization and basic network connectivity troubleshooting.

Software Used: Cisco Packet Tracer